

19TH Experiment

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import make_circles

from sklearn.metrics import accuracy_score

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense


# Generate Circular Dataset

X, y = make_circles(
    n_samples=500,
    noise=0.1,
    factor=0.4,
    random_state=42
)


# Build Neural Network

model = Sequential([
    Dense(1, activation='linear', input_shape=(2,))
])


# Compile Model

model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['accuracy']
)


# Train Model

history = model.fit(
    X, y,
```

```
    epochs=50,  
    verbose=0  
)
```

```
# Plot Training Loss
```

```
plt.figure(figsize=(5,3))  
plt.plot(history.history['loss'])  
plt.title("Training Loss")  
plt.xlabel("Epoch")  
plt.ylabel("Loss")  
plt.grid(True)  
plt.show()
```

```
# Prediction
```

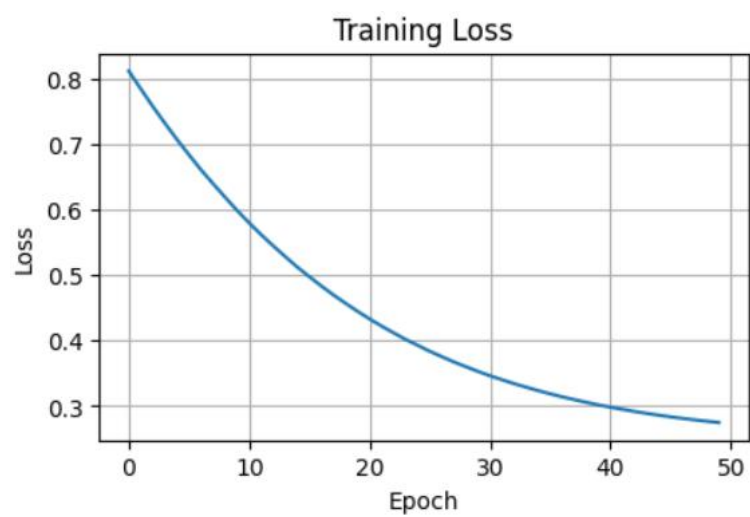
```
pred = model.predict(X)  
pred_class = (pred > 0.5).astype(int)
```

```
# Accuracy
```

```
acc = accuracy_score(y, pred_class)  
print("Accuracy:", round(acc*100,2), "%")
```

```
# Plot Circular Dataset
```

```
plt.figure(figsize=(5,4))  
plt.scatter(X[:,0], X[:,1], c=y)  
plt.title("Circular Data")  
plt.xlabel("Feature 1")  
plt.ylabel("Feature 2")  
plt.show()
```



6/16 ————— 0s 3ms/step
ccuracy: 44.8 %

